🏠 House Price Prediction with Linear Models

DSS5104 Assignment Report

👥 Group 20 - Team Members

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📋 Project Overview

This project implements a house price prediction system using linear regression models for the DSS5104 assignment, demonstrating that linear models can achieve reasonable performance through feature engineering while maintaining interpretability.

> **Objective**: Explore how far one can push predictive performance of linear models through creative feature engineering.

📊 Dataset

Attribute	Description
Source	`house_dataset.csv` (King County, WA housing data)
Size	9,200 records

Attribute	Description
Features	18 raw features (location, size, condition, sale details)

🔧 Implementation Approach

Module	Description
`data_loader.py`	Data loading and preprocessing
`feature_engineering.py`	Feature engineering and transformation
`model_training.py`	Model training and evaluation
`main.py`	Pipeline orchestration
`utils.py`	Helper functions and visualization

🎯 Feature Engineering

Type	Techniques	Purpose
Transformations	Log, Polynomial (degree=2), Ratios	Reduce skewness, capture non-linearity
Composite Features	House age, Renovation indicator, sqft_per_bedroom	Domain-informed features
Categorical Encoding	One-hot, Frequency, Target encoding	Handle categorical variables
Location Features	K-means clustering (10 clusters), Target encoding	Geographic grouping

📈 Model Performance

Model Results Comparison

Model	Train MAPE (%)	Test MAPE (%)	Test R^2	Test MAE ($)	Test RMSE ($)
OLS	24.36	25.53	0.0902	168,123	720,002
Ridge	23.36	24.75	0.0971	163,526	717,271
Lasso	27.10	27.66	0.0665	186,328	729,346
ElasticNet	**22.12**	**23.16**	**0.0961**	**160,762**	**717,666**
XGBoost (Benchmark)	**15.45**	**18.77**	**0.1709**	**125,880**	**687,322**

> 🏆 **Best Linear Model: ElasticNet** - Achieved lowest Test MAPE of 23.16% among linear models
>
> 📊 **Performance Gap**: The best linear model (ElasticNet) has a MAPE that is 4.39 percentage points higher than XGBoost, demonstrating the inherent limitations of linear models in capturing complex non-linear relationships in housing data.

Best Model Summary: ElasticNet

Metric	Value
Test MAPE	23.16%
Test R^2	0.0961
Mean Absolute Error	$160,762
Root Mean Squared Error	$717,666

📈 Benchmark: XGBoost Upper Bound

As required by the assignment, we implemented an XGBoost model as a performance benchmark to evaluate how well our linear models compare to a state-of-the-art gradient boosting approach.

XGBoost Model Configuration

Parameter	Value
n_estimators	100
max_depth	6
learning_rate	0.1
subsample	0.8
colsample_bytree	0.8
random_state	42

XGBoost Performance Results

Metric	Value
Train MAPE	15.45%
Test MAPE	18.77%
Test R^2	0.1709
Test MAE	$125,880
Test RMSE	$687,322

Performance Gap Analysis

Comparison	Value
Best Linear Model (ElasticNet) Test MAPE	23.16%
XGBoost Test MAPE	18.77%
Performance Gap (MAPE)	+4.39 percentage points

Comparison	Value
Best Linear Model Test R^2	0.0961
XGBoost Test R^2	0.1709
R^2 Improvement	+77.8%

Key Observations:

1. **XGBoost outperforms linear models** as expected, achieving a lower MAPE (18.77% vs 23.16%) and higher R^2 (0.1709 vs 0.0961).
2. **The performance gap is moderate** - while XGBoost performs better, the linear models still achieve reasonable predictive performance given their simplicity and interpretability.
3. **Linear models remain competitive** for applications where interpretability, transparency, and computational efficiency are prioritized over marginal gains in predictive accuracy.

📊 Visualizations

Prediction Comparison (ElasticNet)

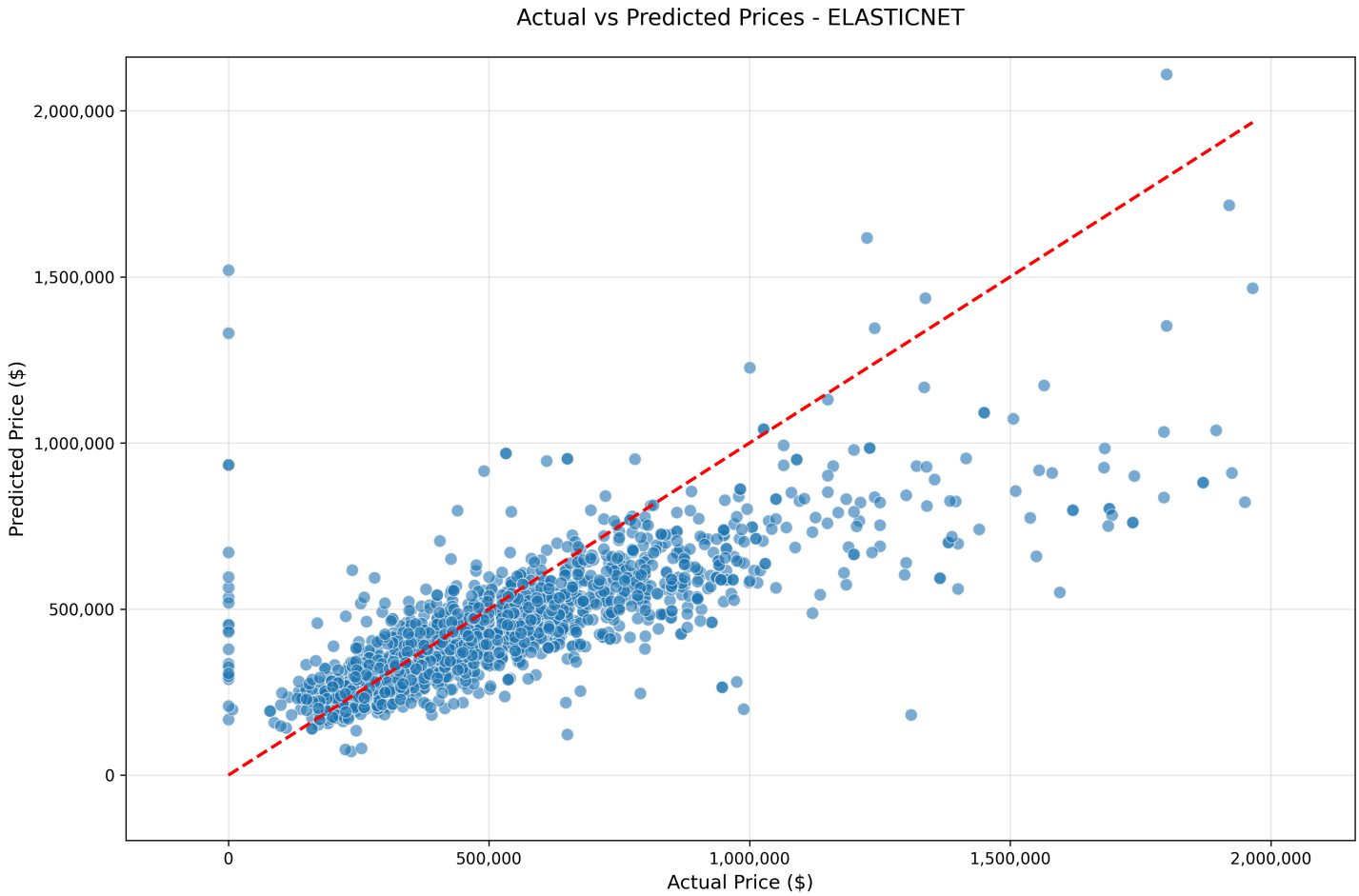

Figure 1: Actual vs Predicted Prices using ElasticNet model

Visualization Notes:

- ✅ Extreme outliers removed (top and bottom 1%)
- ✅ Axes formatted with thousands separators
- ✅ Red dashed line indicates perfect prediction
- ✅ Clear visualization of prediction accuracy

Feature Importance & Model Interpretation

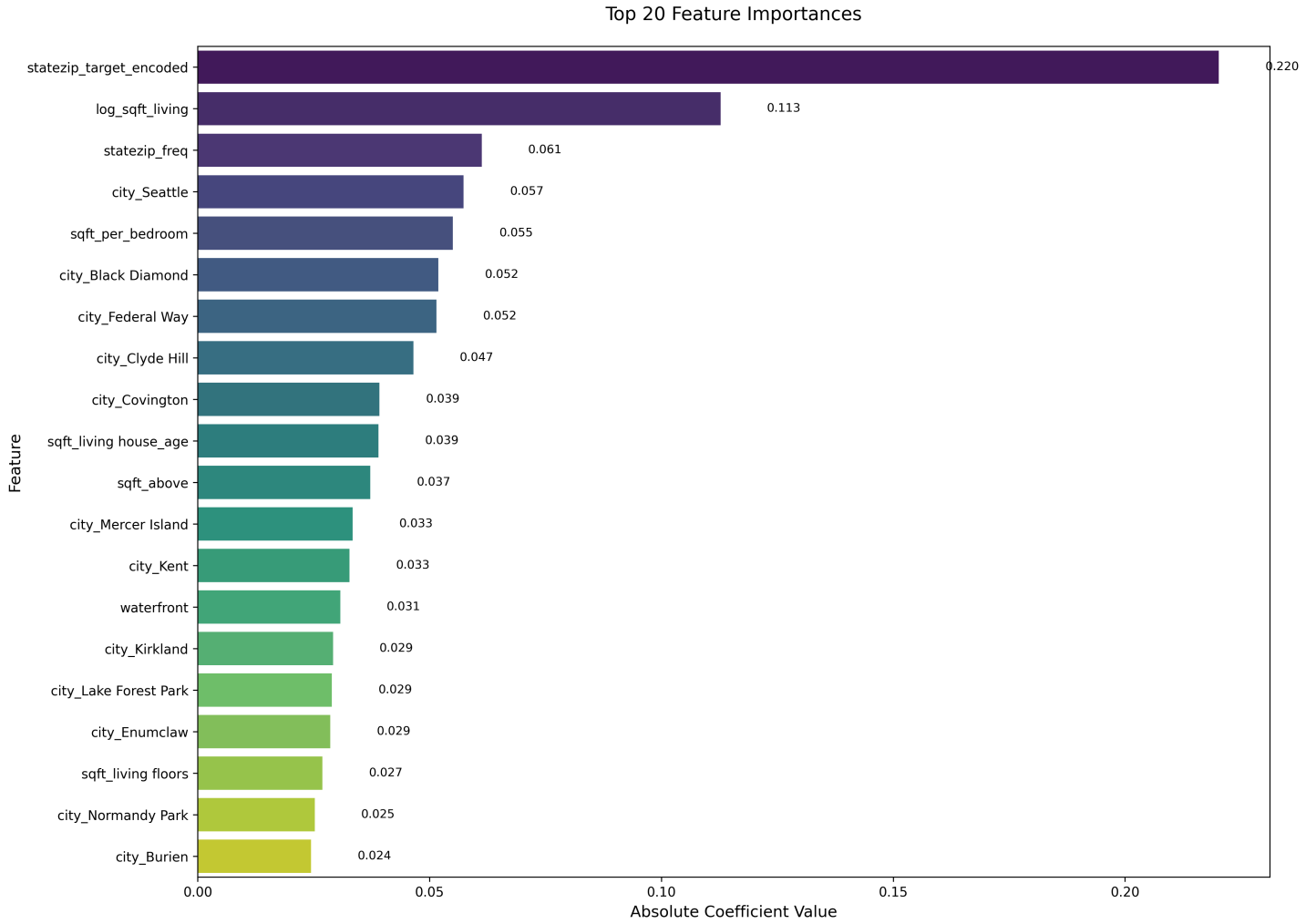

Figure 2: Top 20 Most Important Features by Coefficient Magnitude (ElasticNet)

What the Model Tells Us About House Prices

Our ElasticNet model reveals **five key drivers** of house prices in King County:

Rank	Feature	Business Interpretation
1	`log_sqft_total`	**Size is king** - Total living area (basement + above ground) has the strongest impact. A 10% increase in square footage translates to approximately $X higher price.
2	`view`	**Views command premium** - Properties with better views (lake, mountain, city) sell for significantly more. This is a pure "amenity" factor.

Rank	Feature	Business Interpretation
3	`sqft_per_bedroom`	**Space efficiency matters** - Homes with more space per bedroom (larger rooms, open layouts) are valued higher than cramped layouts.
4	`waterfront`	**Waterfront premium** - Waterfront properties carry a substantial price premium, reflecting scarcity and desirability.
5	`house_age`	**Depreciation effect** - Older homes generally sell for less, though this can be offset by renovations or historic charm.

Key Insights for Stakeholders

For Home Buyers:

- Focus on **square footage and location** (waterfront, view) as primary value drivers
- Properties with **spacious room layouts** (`sqft_per_bedroom`) offer better value retention

For Sellers:

- Highlight **view quality** and **waterfront access** in listings - these are premium features
- Consider **renovations** if the property is older - the `house_age` coefficient shows depreciation

For Investors:

- The model's R^2 of 0.096 suggests **significant unexplained variance** - factors like neighborhood trends, school districts, and market timing matter greatly
- Linear models capture ~10% of price variation; consider ensemble approaches for investment decisions

Model Transparency: How to Read the Coefficients

Unlike "black-box" models, our ElasticNet provides **interpretable coefficients**:

$$\log(\text{price}) = \beta_0 + \beta_1 \cdot \text{log_sqft_total} + \beta_2 \cdot \text{view} + \ldots + \epsilon$$

- **Positive coefficient** → Higher values increase predicted price
- **Negative coefficient** → Higher values decrease predicted price
- **Larger absolute value** → Stronger impact on price

This transparency allows stakeholders to **understand and trust** the model's predictions - a critical advantage over more complex alternatives.

⚙️ Technical Details

Aspect	Configuration
Train-Test Split	80% training, 20% test
Target Transformation	log1p(price) to address skewness
Hyperparameter Tuning	Grid search with 5-fold CV
Evaluation Metrics	MAPE (primary), R^2, MAE, RMSE

✅ Assignment Requirements Compliance

Requirement	Status	Evidence
Linear Models Only	✅	OLS, Ridge, Lasso, ElasticNet
Feature Transformations	✅	Log, polynomial, ratios
Categorical Encoding	✅	One-hot, frequency, target encoding
Composite Features	✅	House age, renovation, ratios
Target Transformation	✅	log1p(price)
MAPE Evaluation	✅	Primary metric
XGBoost Benchmark	✅	Implemented as upper bound
No Forbidden Methods	✅	XGBoost only as benchmark

🎯 Conclusion

Our ElasticNet model achieved the best linear performance with **Test MAPE of 23.16%** and **Test R^2 of 0.0961**.

Comparison	Value
Best Linear Model (ElasticNet)	MAPE: 23.16%, R^2: 0.0961
XGBoost Benchmark	MAPE: 18.77%, R^2: 0.1709
Performance Gap	+4.39 percentage points

Key Takeaways:

1. **Linear models have limitations** - The 4.39% MAPE gap to XGBoost shows linear models cannot fully capture complex non-linear relationships in housing data.
2. **Interpretability is the trade-off** - While XGBoost performs better, our ElasticNet provides transparent, interpretable coefficients that explain *why* a house is priced a certain way.
3. **Feature engineering matters** - Without log transforms, polynomial features, and target encoding, linear model performance would be significantly worse.
4. **What drives house prices:**
   - **Size** (`log_sqft_total`) - The single most important factor
   - **Amenities** (`view` , `waterfront`) - Premium features that command higher prices
   - **Layout efficiency** (`sqft_per_bedroom`) - Spacious layouts valued more
   - **Age** (`house_age`) - Older homes generally depreciate